

John McGahay

Curriculum Vitae

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Education

- 2027 (expected) **Doctor of Philosophy in Linguistics** (advanced to candidacy December 4, 2025)
University of California, Los Angeles | Los Angeles, CA
Dissertation Title: “Confusion Minimization Theory: Towards a Unified Computational Account of Sound Change and Phonological Typology Based on Optimal Listeners and Speakers”
Committee: Bruce Hayes (Co-Chair), Megha Sundara (Co-Chair), David Goldstein, Connor Mayer, Laurel Perkins
- 2024 **Master of Arts in Linguistics**
University of California, Los Angeles | Los Angeles, CA
Thesis Title: “Vowel System Typology and Confusion Minimization”
Committee: Bruce Hayes (Co-Chair), Megha Sundara (Co-Chair), Meg Cychosz
- 2019 **Bachelor of Arts**
University of Pennsylvania, College of Arts and Sciences | Philadelphia, PA
Honors Thesis: “The Effect of Conditioned Mergers on Underlying Representations”
Advised by Prof. Charles Yang
Majors: Linguistics with Honors
Cognitive Science (Language and Mind concentration)
Minors: Computer Science
German
- Spring 2017 **Semester Abroad**
Berlin Consortium for German Studies | Berlin, Germany
German-language coursework at the Freie Universität and Humboldt-Universität
- 2015 **High School Diploma**
Regis High School | New York, NY

Publications

- 2025 McGahay, J. (2025). Modeling Vowel System Typology Using Iterated Confusion Minimization. In *Proceedings of Interspeech 2025* (pp. 2955-2959).
- Under Review McGahay, J. (under review). *A theory of vowel system typology based on confusion minimization. Language.*
- Under Review Cram, C., McGahay, J., & Sundara, M. (under review). *Vowel allophony improves maximum likelihood classification of Warlpiri consonants.* In *Proceedings of Interspeech 2026.*
- In Preparation McGahay, J. (in preparation). *A computational model of allophony as dispersion over phonetic sequences*

Research Presentations

June 2026	Poster Presentation <i>LabPhon 20</i> , Montréal, Québec, Canada “A Computational Account of Merger as Phonetic Dispersion Over Words”
May 2026	Poster Presentation <i>ASA Spring 2026 Annual Meeting</i> , Philadelphia, Pennsylvania, USA “A Bayesian Account of Perceptual Asymmetries in Confusion Matrix Data”
April 2026	Invited Talk UC Berkeley PhonLab, California, USA Tentative title: “A Theory of Vowel System Typology Based on Confusion Minimization”
January 2026	Conference Talk <i>LSA 2026 Annual Meeting</i> , New Orleans, Louisiana, USA “A Computational Model of Allophony as Dispersion over Phonetic Sequences”
August 2025	Conference Talk & Proceedings <i>Interspeech 2025</i> , Rotterdam, The Netherlands “Modeling Vowel System Typology Using Iterated Confusion Minimization”
December 2023	Poster Presentation <i>Acoustics Sydney 2023</i> (co-organized by the ASA), Sydney, Australia “A Theory of Vowel Dispersion Based on Probabilistic Modeling of Optimized Speakers and Listeners”
March 2019	Conference Talk <i>Penn Linguistics Conference 43</i> , Philadelphia, Pennsylvania, USA “The Effect of Conditioned Mergers on Underlying Representations”
Under Review	Conference Talk & Proceedings , with Coralie Cram & Megha Sundara <i>Interspeech 2026</i> , Sydney, Australia “Vowel Allophony Improves Maximum Likelihood Classification of Warlpiri Consonants”

Grants & Fellowships

AY 2025-2026	Graduate Research Mentorship <i>UCLA Graduate Education</i> “Modeling Allophony Using Iterated Confusion Minimization” Advised by Prof. Megha Sundara
Summer 2025	Graduate Summer Research Mentorship <i>UCLA Graduate Education</i> “Modeling the Great Vowel Shift Using Iterated Confusion Minimization” Advised by Prof. Megha Sundara
AY 2024-2025	Graduate Research Mentorship <i>UCLA Graduate Education</i> “Modeling Human Speech Perception Using Automatic Speech Recognition” Advised by Prof. Megha Sundara
Spring 2024	Biggs Graduate Fellowship <i>UCLA Department of Linguistics</i>

- Summer 2023 **Graduate Summer Research Mentorship**
UCLA Graduate Education
 “Testing a New Probabilistic Theory of Phoneme Dispersion”
 Advised by Prof. Bruce Hayes
- Summer 2018 **Jumpstart for Juniors Program for independent summer research**
Penn Center for Undergraduate Research & Fellowships
 “The Effect of Conditioned Mergers on Underlying Representations”
 Advised by Prof. Meredith Tamminga
- Summer 2016 **Penn Undergraduate Research Mentorship**
Penn Center for Undergraduate Research & Fellowships
 Grant for summer RA work for rising second year undergrads, mentored by Prof. Meredith Tamminga

Teaching Experience

- Fall 2024 **Teaching Assistant (50%) to Prof. Laurel Perkins**
UCLA Linguistics Department | Los Angeles, CA
 LING 185A: Computational Linguistics I
 - Running weekly discussion sections and office hours, grading, and responding to student emails
- Spring 2024 **Teaching Assistant (25%) to Prof. Sun-Ah Jun**
UCLA Linguistics Department | Los Angeles, CA
 LING C211: Intonation
 - Running weekly discussion section and office hour, grading, and responding to student emails
- Winter 2024 **Teaching Assistant (50%) to Prof. Sam Zukoff**
UCLA Linguistics Department | Los Angeles, CA
 LING 165A: Phonology II
 - Running weekly discussion sections and office hours, grading, and responding to student emails
- Winter 2024 **Teaching Assistant (25%) to Dr. Elise Bell**
UCLA Linguistics Department | Los Angeles, CA
 LING 102: Introduction to Applied Phonetics
 - Running weekly discussion sections and office hours, and responding to student emails
- Fall 2023 **Teaching Assistant (50%) to Prof. Sun-Ah Jun**
UCLA Linguistics Department | Los Angeles, CA
 LING 103: Introduction to General Phonetics, instructed by Prof. Sun-Ah Jun
 - Running weekly discussion sections and office hours, grading, and responding to student emails
- Sept 2019 – March 2020 **Fulbright English Teaching Assistant**
Nicolaus-Cusanus-Gymnasium | Bonn, Germany
 - Fulbright Scholar in Germany assisting English instruction for middle and high school-aged students
 - Ended three months early due to COVID-19 pandemic
- Sept 2015 – May 2019 **German Tutor**
Penn Language Center | Philadelphia, PA
 - One-on-one tutoring of beginning to intermediate German students at the University of Pennsylvania
- Fall 2018 **Teaching Assistant to Prof. Meredith Tamminga**
Penn Linguistics Department | Philadelphia, PA
 LING 336: Sound Systems in Use (quantitative corpus sociophonetics)
 - Moderating in-class undergraduate discussions, assisting students with R scripting during in-class lab time, holding weekly office hours, responding to student emails

Experience as Research Assistant

- Summer 2024 **Research Assistant to Prof. Jesse Harris**
UCLA Language Processing Lab | Los Angeles, CA
- Splicing and masking of stimuli for pupillometry experiment with phoneme restoration component
- Summer 2023 **Research Assistant to Prof. Sun-Ah Jun**
UCLA Phonetics Lab | Los Angeles, CA
- ToBI-based intonation transcription in experiment stimuli and responses
- May 2016 –
May 2019 **Research Assistant to Prof. Meredith Tamminga**
Penn Language Variation and Cognition Lab | Philadelphia, PA
- Maintenance and Reorganization of the Philadelphia Neighborhood Corpus
 - Processing and analysis of phonetic data using Praat scripting, R, and Python
 - Transcription and forced alignment of sociolinguistic interview with ELAN and FAVE
- Feb 2016 –
May 2016 **Research Assistant to Prof. Jianjing Kuang**
Penn Phonetics Lab | Philadelphia, PA
- Sound segmentation and labeling of Shanghainese wordlist recordings in Praat

Other Work Experience

- June 2021 –
July 2022 **Junior Computational Linguist**
Verbio Technologies | Remote
- Developing intent-classification systems used for call center automation
 - Bash scripting for manipulation of large training datasets
- March 2021 –
June 2021 **Natural Language Understanding Specialist**
Cerence | Remote
- Training and testing of neural-network-based language models used for car voice control features
 - Left due to job insecurity from budget cuts associated with semiconductor shortage
- April 2020 –
Aug 2021 **Computational Linguistics Temporary Worker**
Linguistic Data Consortium | Remote
- Assistant on project to develop a new forced aligner
 - Transcription of clinical and experimental linguistic data for research into early language-based markers of neurodegenerative diseases
- May 2019 –
Aug 2019 **Computational Linguistics Temporary Worker**
Linguistic Data Consortium | Philadelphia, PA
- Phonetic and phonological analysis of Portuguese dialects for speech recognition applications
 - Implementation of grammars in OpenFst-based packages Pynini and Thrax

Awards and Honors

- 2019-2022 **Fulbright Scholar**
Fulbright Germany
- May 2019 **Summa cum Laude**
University of Pennsylvania

May 2019	Major Honors in Linguistics <i>Penn Department of Linguistics</i>
2018-2019, 2017-2018, 2016-2017, 2015-2016	Dean's List <i>University of Pennsylvania</i>
March 2019	Best Undergraduate Abstract <i>Penn Linguistics Conference 43</i>
April 2018	Phi Beta Kappa Inductee <i>Delta Chapter of Phi Beta Kappa at the University of Pennsylvania</i>

Selected Skills

Technical:	Python; Praat (incl. scripting); R; LaTeX; Microsoft Office; Haskell; OCaml; Java; Matlab; ELAN
Language:	English (native); German (fluent); French, Spanish (intermediate, grammatical and phonological familiarity); Irish, Portuguese, Russian, Italian, ASL (basic, grammatical and phonological familiarity); various others (grammatical and phonological familiarity)

References

Bruce Hayes, Ph.D.
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Megha Sundara, Ph.D.
MA & PhD chair
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David Goldstein, Ph.D.
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